

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Pseudo Code Design

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5- Pseudo Code Design	Assessment:	Assessment Tool 1– Pseudo Code Design
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	T.Megha Harsha	Reg. No.:	192424389
Section / Batch:		Date:	

Code of Conduct

I T.Megha Harsha (192424389) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MEGHA

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1.Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.

Aim

To compare kNN, Decision Tree, Logistic Regression, and K-Means models and select the best-performing model.

Algorithm

1. Load the dataset.
2. Split the dataset into training and testing sets.
3. Train kNN, Decision Tree, and Logistic Regression models.
4. Apply K-Means clustering on the dataset.
5. Evaluate models using Accuracy and Silhouette Score.
6. Compare results and select the best model.

Pseudo code:

BEGIN

Load Dataset

Train Models

Evaluate Performance

Compare Results

Select Best Model

Display Result

END

CODE:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.cluster import KMeans
from sklearn.metrics import accuracy_score, silhouette_score

# Load data
X, y = load_iris(return_X_y=True)

# Split data
X_train, X_test, y_train, y_test = train_test_split(
```

```

X, y, test_size=0.2, random_state=42)

# Models
models = {
    "kNN": KNeighborsClassifier(3),
    "Decision Tree": DecisionTreeClassifier(),
    "Logistic Regression": LogisticRegression(max_iter=200)
}

# Train and test
for name, model in models.items():
    model.fit(X_train, y_train)
    pred = model.predict(X_test)
    print(name, "Accuracy =", accuracy_score(y_test, pred))

# K-Means
kmeans = KMeans(n_clusters=3, random_state=42)
clusters = kmeans.fit_predict(X)
print("K-Means Silhouette Score =", silhouette_score(X, clusters))

```

OUTPUT:

```

kNN Accuracy = 1.0
Decision Tree Accuracy = 1.0
Logistic Regression Accuracy = 1.0
K-Means Silhouette Score = 0.55

```

Result:

The model was trained and evaluated successfully. Based on the accuracy value, the model performance was measured and the best model can be selected for the given dataset.